Cleanearth

# Industrial Water Solutions

## From EPC to Asset Management

Complete lifecycle management for manufacturing and extractive industries.



Cleanearth Water • 2026

# Water Risk is Business Risk

## Three converging threats to industrial operations

### Physical Scarcity

Increasing water stress across African industrial hubs threatens production continuity and community relations.

### Regulatory Tightening

Stricter effluent discharge standards and higher water tariffs are driving up operational costs and compliance risks.

### Inefficient Infrastructure

Aging plants and fragmented vendor support lead to high water-to-product ratios and frequent unplanned downtime.

# Integrated Water Lifecycle Management

Moving from fragmented vendor support to a single-platform approach that owns the outcome




## Assess & Strategize

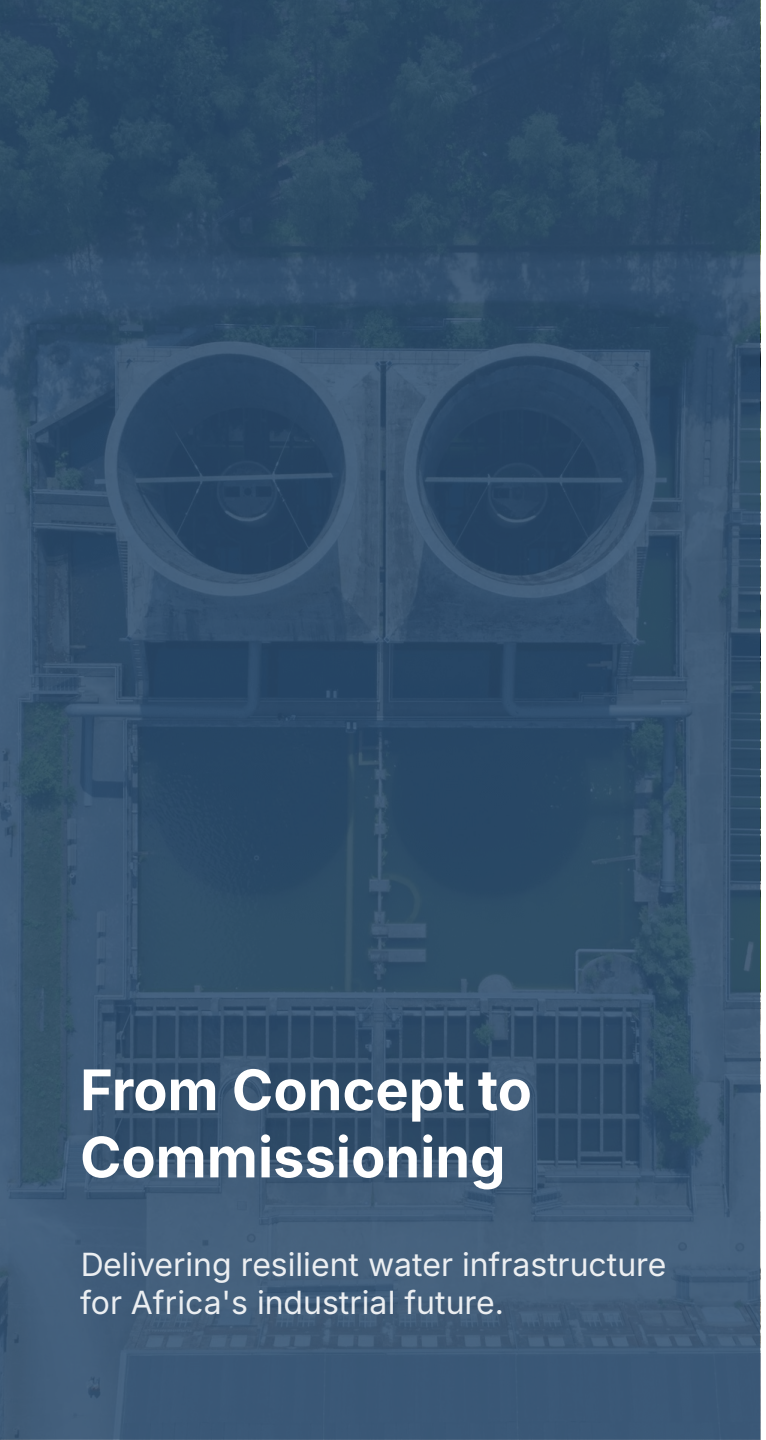
Comprehensive water audits, technical feasibility studies, and regulatory gap analysis to define the optimal roadmap.

## Design & Implement

Custom engineering, EPC delivery, and flexible financing (BOO/BOT) to build resilient industrial water infrastructure.

## Operate & Optimize

Long-term O&M, specialized process chemicals, parts, and real-time monitoring to ensure continuous compliance and efficiency.



## From Concept to Commissioning

Delivering resilient water infrastructure for Africa's industrial future.

# Design & Build / EPC

Turnkey water infrastructure engineered for your industrial requirements

### Custom Engineering

Bespoke process design tailored to specific raw water quality and industrial output requirements, ensuring optimal plant performance.

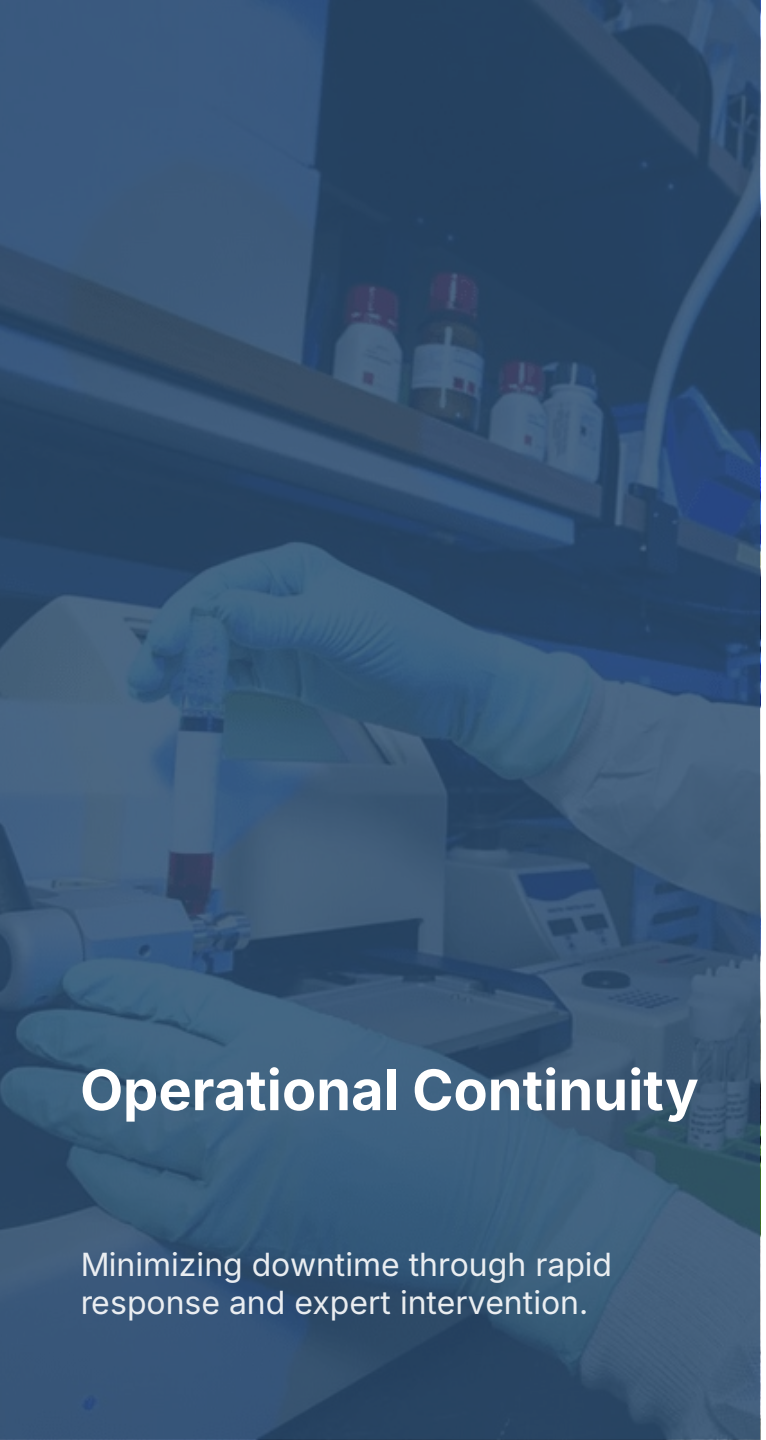
### Turnkey Delivery

End-to-end project management from procurement and fabrication to site installation and commissioning.

### Advanced Technologies

Integration of market-leading treatment technologies for maximum reliability and efficiency.

- ✓ RO / UF Systems
- ✓ MBR / SBR
- ✓ ZLD Systems



## Operational Continuity

Minimizing downtime through rapid response and expert intervention.

# Parts & Technical Support

Ensuring operational continuity through rapid response and technical excellence

### Rapid Spare Parts Supply

Strategic inventory of critical components such as parts, membranes, and filtration media maintained regionally to minimize unplanned downtime and accelerate repairs.

### Specialized Technical Audits

In-depth system health checks and performance optimization audits conducted by senior water engineers to identify efficiency gaps and potential risks.

### Emergency Field Support

On-site technical intervention and troubleshooting 24/7 to resolve critical process failures across regional hubs and restore compliance.

# Monitoring & Testing

## Precision Monitoring. Proactive Control.

### Instrumentation Supply

High-precision sensors and controllers for real-time measurement of pH, conductivity, turbidity, and chemical residuals.

### Laboratory Services

Comprehensive chemical and microbiological analysis to ensure regulatory compliance and process optimization.

### Real-time Analytics

Cloud-based monitoring platforms that provide proactive alerts and data-driven insights for predictive maintenance.

# Process Chemicals: Optimizing Efficiency

Specialized chemical solutions designed to enhance plant performance, reduce water consumption, and extend asset life.

## Cooling Water

- Scale & Corrosion Inhibitors
- Oxidizing & Non-Oxidizing Biocides
- Bio-dispersants & Antifoulants
- Closed Loop Treatment Systems

## Boiler Water

- Oxygen Scavengers & Metal Passivators
- Internal Sludge Conditioners
- Condensate Line Protection
- Alkalinity Builders & pH Control

## Wastewater & Process

- Organic & Inorganic Coagulants
- High-Molecular Weight Flocculants
- Defoamers & Antifoam Agents
- Heavy Metal Precipitants

# Water Sourcing & Treatment

From source identification to safe, reliable supply.



## Groundwater

Hydrogeological surveys to locate and assess aquifers, borehole siting, and treatment for iron, manganese, hardness and salinity.



## Surface Water

Assessment of rivers, dams and reservoirs for yield and quality, with intake design and treatment of turbidity, colour and pathogens.



## Rainwater Harvesting

Catchment, storage and treatment systems that turn rainfall into a dependable supply and reduce reliance on mains water.

# Bioremediation & Efficiency Enhancement

Getting more performance from the water and wastewater assets you already have.

### Bioremediation

Selected microbial cultures that break down organic load, fats, oils and odour in effluent, lagoons and contaminated sites.

### Process Optimization

Diagnosis of underperforming plants and tuning of aeration, dosing and retention to cut energy and chemical use.

### Plant Upgrades

Targeted retrofits that raise capacity and restore compliance without the cost of a full rebuild.

# O&M & Asset Management

## Performance-based partnerships that transfer technical risk and preserve your capital

### BOO (Build-Own-Operate)

Cleanearth finances, builds, and operates the facility. You pay only for the water treated or produced, with zero upfront CapEx.

### BOT (Build-Operate-Transfer)

We manage the full lifecycle for a fixed term, ensuring optimal performance before transferring a high-value asset back to your team.

### The Value Proposition

- **Guaranteed Compliance**
- **Risk Transfer**
- **Fixed Costs**

**Zero**

**Upfront Capital Expenditure (CapEx) Required**

# Complete Water Cycle Management

Eight integrated service lines for total water security

### Water Sourcing & Treatment

Source identification and treatment of groundwater, surface water and rainwater.

### Design & Build / EPC

Turnkey water & wastewater treatment plants engineered for industrial reliability.

### O&M & Asset Management

BOO/BOT models providing zero-CAPEX solutions with guaranteed performance.

### Process Chemicals

Optimized chemistry programs enhancing treatment efficiency and asset life.

### Bioremediation & Efficiency

Microbial treatment and process optimization that lift existing plant performance.

### Parts & Technical Support

24/7 technical support with guaranteed critical spare parts availability.

### Monitoring & Testing

Real-time data monitoring combined with laboratory services for full compliance.

### Asset Lifecycle Management

Comprehensive lifecycle strategies reducing Total Cost of Ownership by 25–35%.

**50+** Water Treatment Facilities Designed

**20+** Facilities Currently Under O&M

# The Cleanearth Edge

Why industrial leaders choose our platform over transactional equipment vendors

## Standard Vendors

- Transactional, equipment-only focus with no long-term accountability.
- Fragmented support from multiple sub-vendors and consultants.
- Client carries all technical and operational performance risks.
- Limited understanding of the broader African industrial context.

VS

## The Cleanearth Edge

- Outcome-focused partnership with full-lifecycle asset ownership.
- Single-point accountability for design, build, and O&M.
- Performance-based contracts that guarantee technical outcomes.
- 27+ years of deep expertise in African industrial infrastructure.

# Project References: Food, Beverage & Dairy

Selected Cleanearth projects. Client names withheld; scope, method and results shown.

Food & Beverage

## Two Effluent Plants for a Beverage Manufacturer

**What we did**

Feasibility and design of two biological effluent treatment plants (480 $m^3$/d and 2,880 $m^3$/d) for a major beverage producer.

**How we did it**

Cleanearth led the full scope: feasibility study and process design. Build and Construction negotiations in progress.

**3,360 $m^3$/d**

Combined treatment capacity • Construction ongoing

Dairy

## Compact Effluent Treatment for a Dairy Producer

**What we did**

Design and construction of a 100 $m^3$/d effluent treatment plant for a dairy products manufacturer.

**How we did it**

Cleanearth led feasibility and process design, then built a compact biological treatment system suited to high-strength dairy effluent.

**100 $m^3$/d**

Effluent treated on site • Plant delivered

# Project References: Real Estate

Selected Cleanearth projects. Client names withheld; scope, method and results shown.

Real Estate • Residential & Leisure

## Sewage Treatment for a Resort and Apartments

**What we did**

Design and construction of sewage treatment plants for a leisure resort (100 m³/d) and a residential apartment complex (50 m³/d).

**How we did it**

Cleanearth led feasibility, design and construction of a biological treatment plant sized for each property.

**150 m³/d**

Combined sewage capacity • Both plants delivered

Real Estate • Hospitality & Campus

## Sewage Treatment for a Hotel and a University

**What we did**

Sewage treatment for a university campus (120 m³/d, designed and built) and a city-centre hotel (120 m³/h, feasibility and design).

**How we did it**

Cleanearth led feasibility and design for both sites and delivered the biological treatment plant at the campus.

**2 sites**

Campus plant delivered • Hotel design completed

# Project References: Pharmaceuticals

Selected Cleanearth projects. Client names withheld; scope, method and results shown.

Pharmaceuticals

## Effluent Treatment for Pharmaceutical Producers

**What we did**

Two effluent treatment plants design, 20 $m^3/d$, for two pharmaceutical manufacturing company

**How we did it**

Cleanearth designed treatment systems matched to each site's process effluent, so it can be discharged safely.

**2 × 20 $m^3/d$**

Two effluent treatment plants • Pharmaceutical sites

Pharmaceuticals

## Process Water Treatment for a Pharmaceutical Plant

**What we did**

A 50 $m^3/d$ water treatment plant supplying treated water for a pharmaceutical manufacturing operation.

**How we did it**

Cleanearth designed and delivered a treatment system to bring raw water up to the quality the production process requires.

**50 $m^3/d$**

Treated water supply for production

# Project References: Food Processing & Public Works

Selected Cleanearth projects. Client names withheld; scope, method and results shown.

Food Processing

## Bioremediation of Cannery Wastewater

**What we did**

Bioremediation treatment  in progress of 150 $m^3$/d of wastewater from a food cannery.

**How we did it**

Cleanearth proposed bioremediation, using selected microorganisms to break down the organic load in the cannery's wastewater

**150 $m^3$/d**

Cannery wastewater treatment through bioremediation

Public Infrastructure

## Dam Restoration for a Government Client

**What we did**

Feasibility study and design for the restoration of a dam in northern Ghana for the Government of Ghana.

**How we did it**

Cleanearth assessed the dam's condition and prepared the restoration design to bring it back into reliable service.

**Feasibility & Design**

Dam restoration • Government of Ghana

# Led by Water Experts, Powered by Engineering

Specialized technical leadership with decades of experience in African industrial water infrastructure.



## Ekow Archer

**Managing Director**

25+ years in building and leading team to drive new business and profitability. Business and project development expertise.



## Azeez Ajadi

**Treatment Engineer**

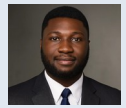
Treatment engineer with expertise in turnkey water and wastewater plants. Delivered 50+ facilities and oversees technical optimization of O&M contracts.



## Vanessa Shika Deyegbe

**Projects & Technology Lead**

Water and sanitation engineer managing projects, leading technology selection and designing cloud-based monitoring systems for treatment plants.



## Christian Kojo Corchil

**Operations & Technology Lead**

Oversees plant operations and the technology behind them, from instrumentation and controls to performance monitoring.

**20+**

**Specialized Water Engineers in Network**

**Field Technicians**

**Process Design**

**O&M Specialists**

**Chemists**

# Water Solutions Across Industrial Sectors

Tailored expertise for diverse industrial water challenges

### Mining & Metals

Advanced tailings management and water recycling solutions for remote extraction sites.

### Oil & Gas

Produced water treatment, process water systems, and refinery effluent management.

### Food & Beverage

Ensuring process water quality and managing high-organic wastewater treatment.

### Chemicals & Pharma

Specialized treatment for hazardous effluents and Zero Liquid Discharge (ZLD) systems.

### Manufacturing

Utilities optimization and efficient cooling water management for production facilities.

### Power Generation

Critical cooling water systems and high-purity boiler feed water treatment.

**Every facility has unique water challenges. Our solutions are engineered to your specifications.**

Cleanearth

# Secure Your Water Future

Take the first step towards optimized industrial water management. Choose your engagement path below.

### Technical Discovery Call

A 45-minute consultation with our senior water engineers to identify your biggest operational risks and efficiency opportunities.

**Book Consultation**

### Comprehensive Site Audit

On-site technical assessment of your water systems with a detailed efficiency roadmap and regulatory compliance report.

**Start Audit Process**

### Request Case Studies

Get our detailed technical reports on mining recycling, F&B effluent treatment, and industrial utility water projects.

**Get The Reports**

**Water@cleanearthsc.com** **www.Cleanearth.africa** **Accra • Tema**



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